

Booking Core — Local Setup

Healthcare appointment booking system (Milestone 1: the reliability spine). Stack: **Node.js** · **Express** · **PostgreSQL** · **Prisma** · **pg-boss**, deployed on Railway/Render.

Status: Milestone 1.1 (foundation & data layer) is code-complete. The booking API, worker, and payments arrive in M1.2 / M1.3 — see [../MILESTONES.md](#).

Prerequisites

- Node.js LTS (v20 or v22)
- Docker (for local Postgres) — or any PostgreSQL 15+ instance
- npm

One-time setup

```
# 1. Start Postgres (Local Docker)
docker compose up -d

# 2. Configure environment
cp .env.example .env          # then edit values as needed

# 3. Install dependencies
npm install

# 4. Generate the Prisma client + apply migrations
npx prisma generate
npx prisma migrate deploy    # applies init + hard_constraints (needs btree_gist)

# 5. Seed a demo doctor, templates, and materialized slots
npm run seed
```

Everyday commands

Command	What it does
<code>npm test</code>	Run the unit test suite (state machine + slot math)
<code>npm run test:integration</code>	Prisma-based integration tests (needs a running DB)

Command	What it does
<code>npm run verify:locking</code>	Prove the no-double-booking guarantee (50×20 concurrency rounds)
<code>npm run verify:expiry</code>	Prove expired reservations release and paid ones never do
<code>npm run seed</code>	(Re)seed the demo clinic and materialize slots
<code>npx prisma studio</code>	Browse the database in a GUI
<code>npm run build</code>	Compile TypeScript to <code>dist/</code>
<code>npm run dev</code>	Run the web server (API) with hot reload
<code>npm run worker</code>	Run the background worker (expiry + sweep + materialization)

What exists today (M1.1)

- **Schema & migrations** — all tables, indexes, and the two hard DB constraints (`one_live_slot_hold`, `no_overlap_per_doctor`).
- **State machine** — `src/domain/stateMachine.ts`, the pure safety net for all slot transitions.
- **Slot materialization** — `src/slots/` turns availability templates into concrete, lockable `AppointmentSlot` rows.
- **Plumbing** — env validation, logging, Prisma client, event logging.

Verifying the constraints (after `migrate deploy`)

```
-- The GiST exclusion constraint should reject an overlapping slot for a doctor:
\d "AppointmentSlot"          -- shows no_overlap_per_doctor
-- The partial unique index guarding live holds:
\di one_live_slot_hold
```

A full `SCHEMA.md`, `API.md`, `WEBHOOKS.md`, `DEPLOYMENT.md`, and the Postman collection are M1.3 deliverables.